

Sales Dashboard Analysis Using Power BI

Certificate

This is to certify that the project entitled "**Sales Dashboard Analysis Using Power BI**" has been carried out by the student as a part of the Final Year Project requirement for the award of the degree. The work presented in this report is original and has been completed under the guidance of the project supervisor.

Declaration

I hereby declare that the project report titled "**Sales Dashboard Analysis Using Power BI**" is my original work and has not been submitted to any other institution for the award of any degree or diploma.

Student Signature

Acknowledgement

I express my sincere gratitude to my project guide, faculty members, and institution for their valuable guidance and support throughout the project. I also thank my friends and family for their encouragement and assistance during the completion of this project.

Abstract

Business Intelligence tools play a vital role in transforming raw business data into meaningful insights. This project focuses on the development of an interactive **Sales Dashboard using Power BI**. The dashboard provides a comprehensive analysis of sales performance across different countries, regions, and sales channels.

The dashboard enables users to monitor key performance indicators such as Total Revenue, Total Cost, Total Profit, and Units Sold. Various visualizations

including KPI cards, bar charts, waterfall charts, line charts, and slicers are used to present business information effectively. The project helps organizations make data-driven decisions by identifying profitable markets, analyzing sales trends, and evaluating regional performance.

Chapter 1: Introduction

1.1 Background

In today's competitive business environment, organizations generate large amounts of sales data. Analyzing this data manually is time-consuming and often leads to inaccurate decision-making. Business Intelligence tools such as Power BI provide interactive visualizations that help organizations gain insights from their data.

1.2 Problem Statement

Organizations often face challenges in:

- Tracking sales performance.
- Comparing revenue and costs.
- Identifying profitable regions.
- Monitoring sales channels.
- Making informed business decisions.

To address these challenges, a Power BI Sales Dashboard was developed.

1.3 Objectives

The objectives of the project are:

1. To analyze sales performance using Power BI.
2. To compare revenue, cost, and profit.
3. To identify top-performing countries and regions.

4. To monitor sales channel performance.
 5. To provide interactive business insights for decision-making.
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Chapter 2: Literature Review

Business Intelligence (BI) refers to technologies and practices used for collecting, integrating, analyzing, and presenting business information.

Power BI is a business analytics tool developed by Microsoft that enables users to:

- Connect to multiple data sources.
- Transform and clean data.
- Create data models.
- Build interactive dashboards.
- Generate business reports.

Many organizations use Power BI for sales analysis because it provides real-time insights and advanced visualization capabilities.

Chapter 3: System Analysis

Existing System

Traditional reporting systems depend on spreadsheets and manual calculations.

Limitations

- Time-consuming
- Lack of interactivity
- Difficult to identify trends
- Increased possibility of errors

Proposed System

The proposed system uses Power BI to create an interactive dashboard that allows users to:

- View sales performance instantly.
- Filter data by country.
- Analyze regional performance.
- Compare revenue, cost, and profit.

Advantages

- Faster reporting
 - Interactive analysis
 - Better decision-making
 - Improved data visualization
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Chapter 4: Dataset Description

The project uses a sales dataset containing information related to global sales transactions.

Dataset Attributes

Attribute	Description
Region	Sales region
Country	Country name
Item Type	Product category
Sales Channel	Online/Offline

Attribute	Description
Order Priority	Priority level
Order Date	Date of order
Order ID	Unique order number
Ship Date	Shipping date
Units Sold	Number of units sold
Unit Price	Selling price per unit
Unit Cost	Cost per unit
Total Revenue	Total sales revenue
Total Cost	Total sales cost
Total Profit	Profit generated

Chapter 5: System Design

Data Flow

Sales Dataset → Data Cleaning → Data Modeling → Power BI Dashboard → Business Insights

Dashboard Components

KPI Cards

- Average Total Cost
- Maximum Units Sold

Gauge Chart

Displays Total Revenue performance against the target.

Bar Chart

Compares Total Revenue and Total Cost.

Waterfall Chart

Shows Unit Price contribution by Region and Sales Channel.

Funnel Chart

Displays Revenue, Cost, and Profit comparison.

Line Chart

Shows Revenue generated by Country.

Slicer

Country-based filtering for interactive analysis.

Chapter 6: Dashboard Analysis

Key Performance Indicators

Average Total Cost

931.81K

This KPI indicates the average cost incurred per transaction.

Total Revenue

137.35 Million

The organization generated a total revenue of 137.35 million.

Total Cost

93.18 Million

The total operational and product cost amounted to 93.18 million.

Total Profit

44.17 Million

The organization achieved a profit of 44.17 million.

Maximum Units Sold

9,925 Units

The highest number of units sold in a single transaction was 9,925.

Regional Analysis

The dashboard indicates that:

- Sub-Saharan Africa recorded the highest units sold.
 - Europe ranked second in sales volume.
 - Asia and Middle East regions contributed significantly.
 - North America recorded comparatively lower sales.
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Sales Channel Analysis

The Waterfall Chart demonstrates the contribution of:

- Online Sales
- Offline Sales

Both channels contribute significantly to overall revenue generation.

Country Analysis

Revenue analysis by country helps identify:

- High-performing countries
- Low-performing countries
- Revenue distribution across global markets

This enables management to focus on profitable regions.

Chapter 7: Results and Findings

The major findings of the project are:

1. Total Revenue generated is **137.35 Million**.
 2. Total Profit earned is **44.17 Million**.
 3. Sub-Saharan Africa has the highest sales volume.
 4. Revenue significantly exceeds total cost, indicating profitability.
 5. Online and Offline channels both contribute effectively to sales.
 6. Country-wise analysis helps identify target markets.
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Chapter 8: Benefits of the System

The dashboard provides the following benefits:

- Real-time sales monitoring
 - Improved business decision-making
 - Easy identification of profitable regions
 - Better visualization of business performance
 - Enhanced reporting efficiency
 - Interactive data exploration
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Chapter 9: Future Enhancements

Future improvements can include:

- Integration with live databases
- Predictive sales forecasting using Machine Learning
- Customer segmentation analysis
- Product performance analytics
- Mobile dashboard deployment

- Real-time sales monitoring
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Chapter 10: Conclusion

The **Sales Dashboard Analysis Using Power BI** project successfully demonstrates the application of Business Intelligence techniques in sales analytics. The dashboard transforms raw sales data into meaningful insights through interactive visualizations and KPI monitoring.

The analysis shows that the organization generated substantial revenue and profit while maintaining effective cost management. The dashboard enables management to monitor performance across countries, regions, and sales channels, thereby supporting data-driven decision-making.

Overall, the project highlights the importance of Power BI as a powerful tool for business analysis and strategic planning.

References

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2. Microsoft Learn – Power BI Training.
3. Business Intelligence Concepts and Applications.
4. Data Visualization Best Practices.
5. Sales Analytics and Reporting Techniques.